

Lutalyse™

Dinoprost tromethamine

zoetis

Each mL contains 5 mg dinoprost (as dinoprost tromethamine) and 16.5 mg benzyl alcohol as the preservative.

Product Description: It is a colorless solution.

Pharmacodynamics: The therapeutic use of dinoprost (natural prostaglandin $F_{2\alpha}$) in food producing animals is to induce regression of corpora lutea. This response is observed in many animals. The intracellular mechanism by which dinoprost induces luteolysis remains unknown. Various other physiological responses to dinoprost are known. These include stimulation of smooth muscle including vascular, bronchial, uterine and gastrointestinal muscle.

Pharmacokinetics: Following administration, dinoprost tromethamine is rapidly dissociated to dinoprost (PGF_{2a}). This compound has an extremely short blood half-life of only a few minutes. Near complete clearance occurs on one to two passages through the liver or lungs. No accumulation of dinoprost or residues has been observed in blood following repeated daily injections in cattle. Highest tissue concentrations of dinoprost are observed at the injection site which deplete to background concentrations by 24-48 hr post-injection. Residue concentrations in milk of cows peak at 2 hr post-injection and decrease rapidly thereafter.

Indication: Lutalyse is indicated for its luteolytic and/or oxytocic effects in cattle, horses and pigs. The indications for use are:
Cattle

1. To more effectively control the time of oestrus in cycling cows.
2. To treat cows which have a functional corpus luteum, but do not express behavioural oestrus (sub-oestrus or silent heat).
3. To induce abortion.
4. To induce parturition.
5. For treatment of chronic metritis and pyometra.
6. For controlled breeding (oestrus synchronisation).

Mares

1. To more effectively control the time of oestrus in cycling mares.
2. To treat mares which have a functional corpus luteum but do not express behavioural oestrus.
3. To induce abortion.

Pigs

1. Induction of parturition
2. To reduce the weaning to oestrus interval (WOI) and the weaning to fertile service interval (WFSI) in sows in herds with reproductive problems.

Recommended Dose:

A single injection of the recommended dose of dinoprost activity is luteolytic provided a functional corpus luteum is present. Administer by intramuscular injection. Full aseptic precautions must be taken. Use a sterile syringe and needle and make the injection through an area of clean dry skin. Care should be taken to avoid injection through wet or dirty areas of skin.

Cattle:

The dose for all indicated uses in cattle is 5 ml Lutalyse (25 mg dinoprost)

Mares:

The dose for all indicated uses in mares is 1 ml Lutalyse (5 mg dinoprost)

Pigs:

1. Induction of parturition: 2 ml Lutalyse (10 mg dinoprost) within 72 hours of expected parturition.
2. Use post partum: 2 ml Lutalyse (10 mg dinoprost) 24 to 48 hours after parturition.

Route of Administration: Administer by intramuscular injection.

Contraindication: Contra-indicated in animals suffering from either acute or sub-acute disorders of the vascular system, gastro-intestinal tract or respiratory system. Contra-indicated when abortion or parturition is not intended. Do not administer by the intravenous route.

Warning and Precautions:

Warnings:

Localized post injection bacterial infections that may become generalized have been reported. Aggressive antibiotic therapy, particularly covering clostridial species, should be employed at the first sign of infection. Careful aseptic techniques should be employed to decrease the possibility of post injection bacterial infections.

Precautions:

1. Prostaglandins of the F_{2a} type can be absorbed through the skin and may cause bronchospasm or miscarriage.
2. Care should be taken when handling the product to AVOID SELF-INJECTION or SKIN CONTACT.

3. Women of child-bearing age, asthmatics and persons with bronchial or other respiratory problems should avoid contact with, or wear disposable plastic gloves, when administering the product.
4. Accidental spillage on the skin should be washed off immediately with soap and water.
5. Animals should not be treated if they suffer from either acute or sub-acute disorders of the vascular system, gastro-intestinal tract or respiratory system.
6. Pregnancy status should be determined prior to injection since Lutalyse has been demonstrated to result in abortion or parturition induction when administered at sufficiently high doses to many animal species.
7. Lutalyse is ineffective when administered prior to day five after ovulation.
8. If pregnant, the unlikely possibility of uterine rupture should be borne in mind, especially if cervical dilation does not occur.
9. Induction of parturition in pigs earlier than 72 hours prior to predicted farrowing date may result in reduced piglet viability.

Interaction with Other Medicinal Product: None known.

Pregnancy and Lactation: Termination of pregnancy is an indication for use.

Side Effects:

Cattle

The most frequently observed side-effect is increased rectal temperature at 5 to 10 times the recommended dose. However, rectal temperature changes have been transient in all cases observed and have not been detrimental to the animal. Limited salivation has been seen in some instances. Localized post injection bacterial infections that may become generalized have been reported. Aggressive antibiotic therapy, particularly covering clostridial species, should be employed at the first sign of infection. Careful aseptic techniques should be employed to decrease the possibility of post injection bacterial infections.

Horses

The most frequently observed side-effects are sweating and decreased rectal temperatures. However, these have been transient in all cases observed and have not been detrimental to the animal. Other reactions seen have been increase in heart rate, increase in respiration rate, some abdominal discomfort, locomotor incoordination and lying down. These effects are usually seen within 15 minutes of injection and disappear within one hour. Mares usually continue to eat during the period of expression of side-effects.

Pigs

Transient side-effects consisting of increased body temperature, increased respiratory rate, increased salivation, stimulation of defaecation and urination, flushing of the skin and restlessness (arching of back, pawing, and rubbing and gnawing the crate) occur occasionally following the administration of dinoprost in pregnant sows and gilts. These effects tend to parallel the signs exhibited by sows prior to normal parturition, only they appear to be condensed in time. These effects are usually seen within 15 minutes of injection and disappear within one hour.

Overdose and Treatment: Seek the advice from licensed veterinarian.

Incompatibilities: None known.

Storage Condition: Store below 30°C.

Following withdrawal of the first dose, the product should be used within 28 days.

Withdrawal Period:

Pigs (meat & offal): 1 day

Cattle (meat & offal): 1 day

Cattle (milk): Zero hours Horses: Not to be used in horses intended for human consumption.

Treated horses may never be slaughtered for human consumption. The horse must have been declared as not intended for human consumption under national horse passport legislation.

How supplied: LUTALYSE is available in 30ml vials.

Manufactured by:

Zoetis Belgium SA

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Product Registration Holder:

Zoetis Malaysia Sdn. Bhd.

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